

An Introduction to Economics and Engineering Economics

UNIT - I

WHAT IS ECONOMICS?

Economic theory explains the laws and principles that govern how an economy and its parts function. An economy exists because of **two basic facts**.

01

Unlimited Wants

Human wants for goods and services are virtually endless.

02

Scarce Resources

The productive resources used to make them are strictly limited.

The Problem of Scarcity

Because wants are unlimited but resources scarce, every society must choose how to allocate its resources to achieve the greatest possible satisfaction of its members.

SCARCITY & CHOICE

Scarce resources force every society to make hard choices about what to produce.

01

Limited Resources

Land, raw materials, skilled manpower and capital exist only in limited quantity.

02

The Problem of Choice

With wants unlimited, society must choose which wants to satisfy first.

03

Trade-offs

Using more resources in one line of production means withdrawing them from another.

Getting the Most from What We Have

Scarcity compels us to allocate resources across competing uses to achieve the greatest possible satisfaction of people's wants.

DEFINITION

Economists across generations have defined economics differently — three definitions stand out.

Adam Smith

1776

“An inquiry into the nature and causes of the wealth of nations.”

Alfred Marshall

1922

“The study of mankind in the ordinary business of life and its material requisites of well-being.”

Lionel Robbins

1935

“The science which studies human behaviour as a relationship between ends and scarce means that have alternative uses.”

WHAT IS ENGINEERING ECONOMICS?

Engineering economics bridges economic theory and real-world business decisions.

01

What It Is

The integration of economic theories and tools of analysis with the decision-making process of business management.

02

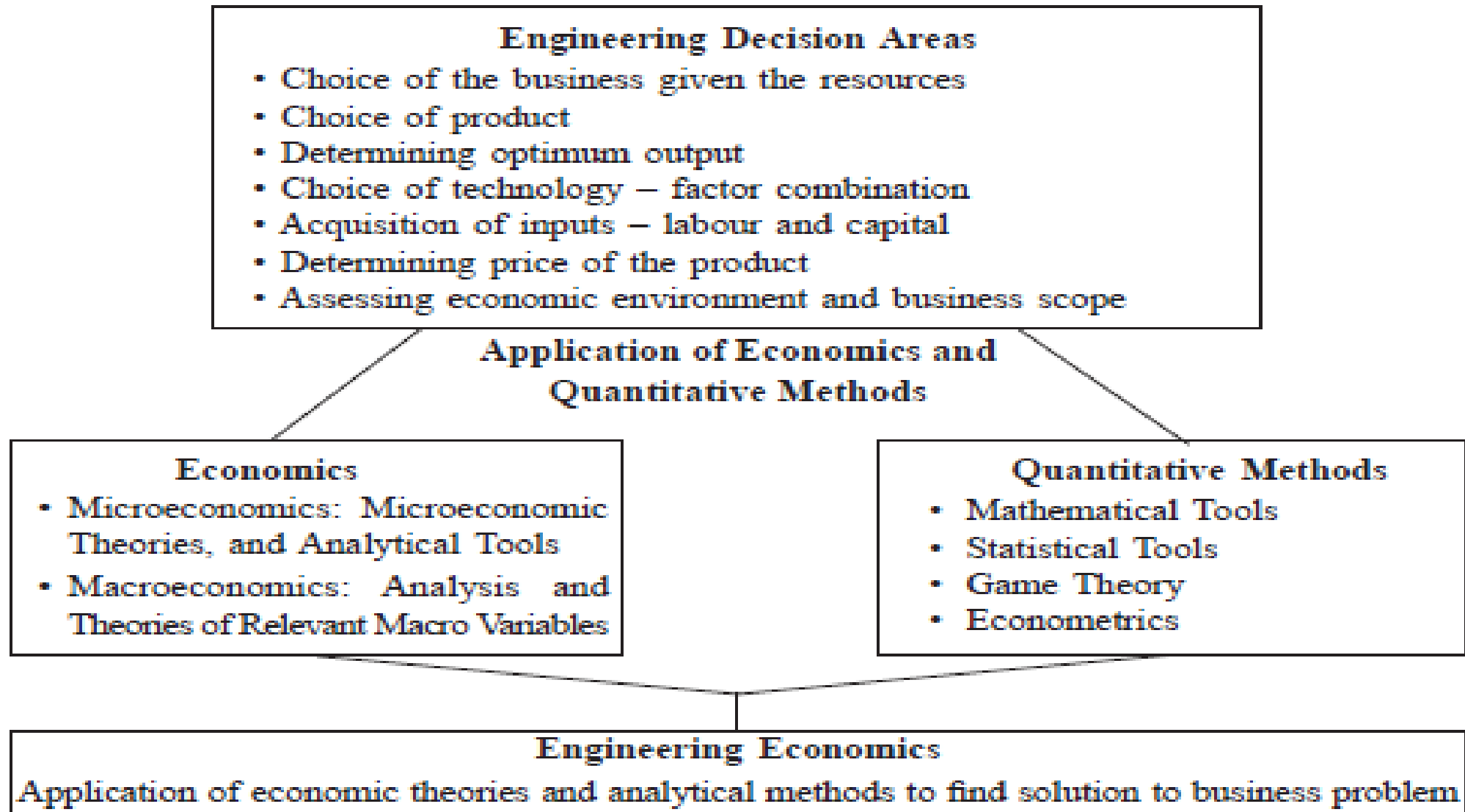
Why It Matters

Applying economic theories and tools to business decisions provides guidance to arrive at appropriate managerial decisions.

ENGINEERING ECONOMICS

Scope of Engineering Economics





Application of Economics in Decision-Making Process: An Illustration

1: Determining and Defining Objective: The Basic Function of Decision-Maker

2: Identifying Business Related Issues: Looking at Market Conditions

3: Collection and Analysis of Relevant Market Data

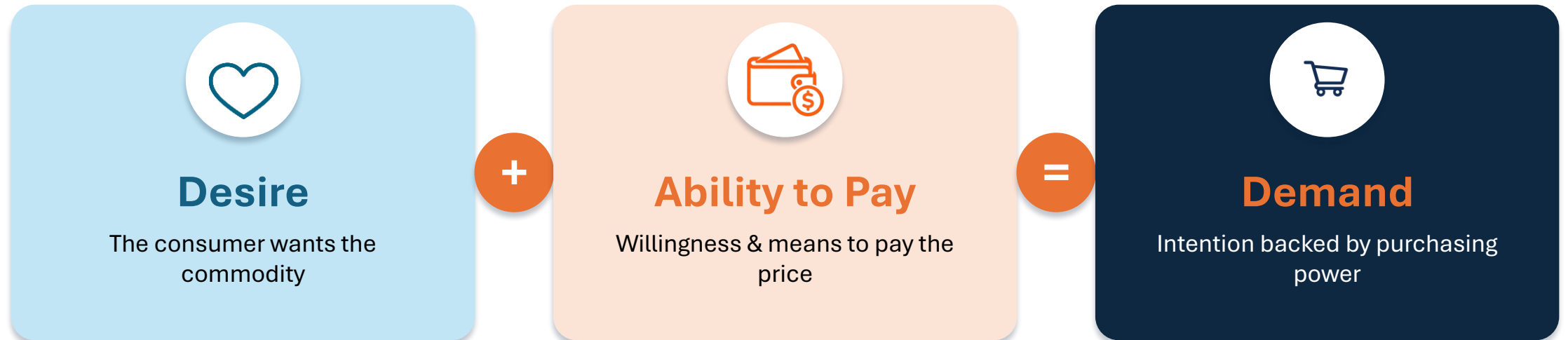
4: Inventing/Deriving Possible Course of Actions: Analysing Market Data

5: Selection of the Best Solution

6: Implementation of the Decision

Definition of Demand

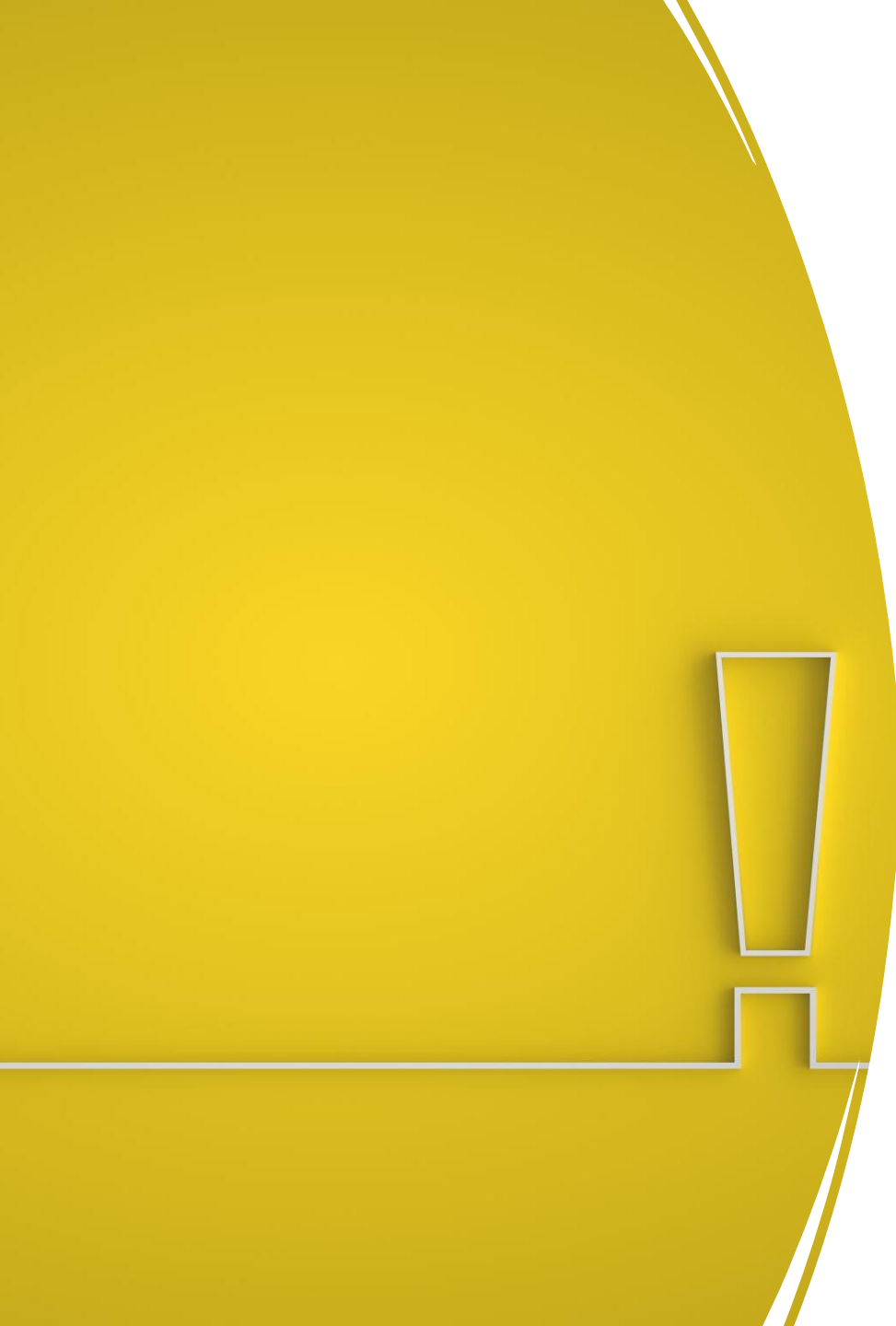
Demand rests on two conditions working together:



Formal definition

The amount of a commodity a consumer will purchase at various given prices during a period of time.

- **It is noteworthy that a mere desire for a commodity does not constitute a demand for it if it is not backed by the ability to pay.**
- Demand for a good is determined by several factors, such as the price of a commodity, the tastes and desires of the consumer for a commodity, the income of the consumer, the prices of related goods, substitutes or complements.
- When there is a change in any of these factors, the demand of the consumer for a good changes.
- Individual consumer's demand and market demand for a good may be distinguished.
- Market demand for a good is the total sum of the demands of individual consumers who purchase the commodity in the market.



DEMAND AND UTILITY

- People demand goods because they satisfy the wants of the people.
- The utility means the amount of satisfaction which an individual derives from consuming a commodity.
- It is also defined as the **want-satisfying power of a commodity**.
- Utility of a good is an important determinant of consumer demand.
- It is assumed that Individuals will try to maximize their utility or satisfaction from the goods they buy for consumption.

DEMAND AND QUANTITY DEMANDED

- **Demand** refers to the quantities of a commodity which consumers plan to buy at various prices of a good during a period of time, while the **quantity demanded** is the amount of a good or service which consumers plan to buy at a particular price.
- It is worth mentioning that the quantity demanded is not necessarily the amount *actually bought by* the consumers.
- Quantity demanded is a *flow concept*.
- This means that quantity demanded is measured as the amount that consumers wish to buy per unit of time, which may be a day, a week, a month, or a year.
- Thus, if you consume 5 cups of tea in a day, then your quantity demanded of tea is expressed as 5 cups per day or 35 cups per week or 150 cups per month, and so on.

DEMAND FUNCTION

- The demand function for a commodity describes the relationship between the quantity demanded of it and the factors that influence it.
- Individual's demand for a commodity depends on its own price, his income, prices of related commodities (which may be either substitutes or complements), his tastes and preferences, and advertising expenditure made by the producers for the commodity in question.
- Individual demand function for a commodity can be expressed in the following general *form* :

$$Q_d = f(P_x, I, P_r, T, A)$$

- P_x = Own price of the commodity X
- I = Income of the individual
- P_r = Prices of related commodities
- T = Tastes and preferences of the individual consumer
- A = Advertising expenditure made by the producers of the commodity.

- For many purposes in economics, it is useful to focus on the relationship between quantity demanded of a good and its own price while ***keeping other determining factors such as consumer's income, prices of other goods, tastes and preferences constant.***

$$Q_d = f(P_x)$$

- There is an inverse relationship between the price of a commodity and its quantity demanded. Thus, when the price of a commodity falls, its quantity demanded will increase, and when its price rises, its quantity demanded will decrease.
- Therefore, when we express this relationship through a curve, we get a downward-sloping demand curve of a commodity.

- Generally, demand function is considered to be of a *linear* form. The specific demand function of a linear form is written as: $Q_d = a - bP$
- where a is a constant intercept term on the X - axis and b is the coefficient showing the slope of the demand curve. The coefficient of price (P), that is, b being negative, implies that there is a negative relationship between the price and quantity demanded of a commodity.
- **Example:** If sugar is demanded at its various prices by an individual consumer is provided, and we find the constant a (*intercept*) to be equal to 70 and the constant b to be equal to 5, we can write the individual's demand function as:

$$Q_d = 70 - 5P$$

- The above demand equation is interpreted as one rupee fall in price of sugar will cause its quantity demanded to increase by 5 units of sugar and vice-versa.

LAW OF DEMAND

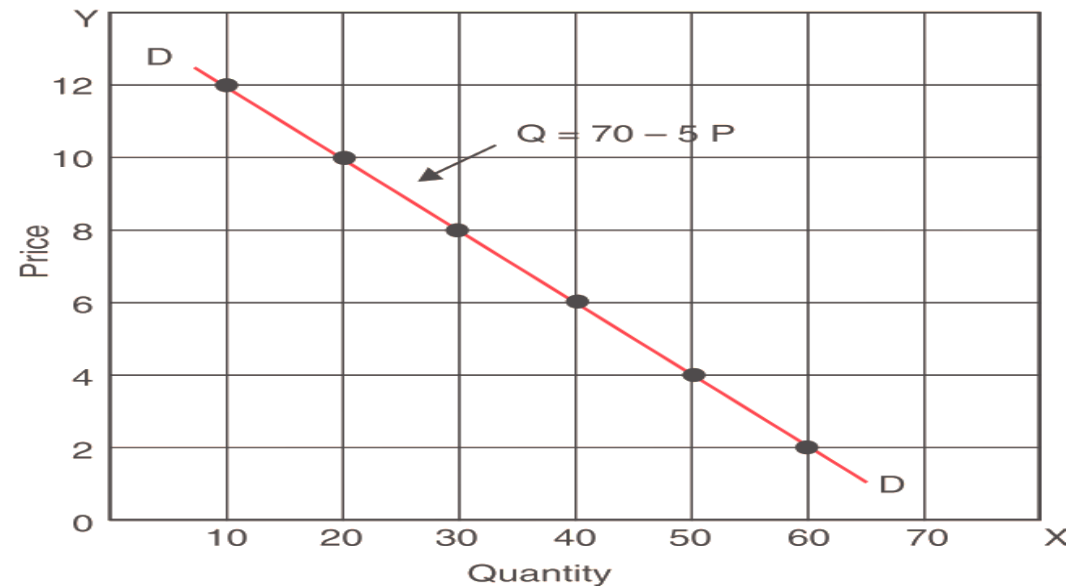
- *According to the law of demand, other things being equal, if the price of a commodity falls, the quantity demanded of it will rise, and if the price of the commodity rises, its quantity demanded will decline.*
- These other things which are assumed to be constant are the tastes and preferences of the consumer, the income of the consumer, and the prices of related goods.
- If these other factors which determine demand also undergo a change at the same time, then the inverse price-demand relationship may not hold good.
- Thus, the constancy of these other things, generally stated as **ceteris paribus**, is an important qualification of the law of demand.
- The law of demand can be illustrated through a demand schedule and a demand curve.
- *The downward-sloping demand curve is in accordance with the law of demand which, as stated above, describes an inverse price-demand relationship.*

- It should be noted that when there is a change in the other determining factors that are held constant, such as income, tastes, and prices of related commodities, the whole demand curve will shift.
- For example, if income increases, the whole demand curve will shift to the right, and, on the contrary, if income decreases, the whole demand curve will shift to the left.
- Similarly, changes in other determining factors such as tastes, prices of related commodities, and advertising expenditure cause shifts in the demand curve and are called *shift factors*.
- If any change occurs in any of these other factors, the whole demand schedule or demand curve will change, and a new demand schedule or a demand curve will have to be drawn.

Demand Schedule of an Individual Consumer

Price (Rs.)	Quantity Demanded
12	10
10	20
8	30
6	40
4	50
2	60

The Demand Curve of an Individual



Reasons for the Law of Demand: Why does Demand Curve Slope Downward?

Income Effect

- When the price of a commodity falls, the consumer can buy more quantity of the commodity with his *given income*. Or, if s/he chooses to buy the same amount of quantity as before, some money will be left with her/him because s/he has to spend less on the commodity due to its lower price.
- In other words, as a result of a fall in the price of a commodity, a consumer's real income or purchasing power increases. This increase in real income induces the consumer to buy more of that commodity.
- This is called the **income effect** of the change in the commodity's price. This is one reason why a consumer buys more of a commodity when its price falls.

Substitution Effect

- When the price of a commodity falls, it becomes relatively cheaper than other commodities.
- This induces the consumer to substitute the commodity whose price has fallen for other commodities which have now become relatively dearer.
- As a result of this **substitution effect**, the quantity demanded of the commodity, whose price has fallen, rises.

Exceptions to the Law of Demand

Goods having Prestige Value: Veblen Effect

- One exception to the law of demand is associated with the name of the economist Thorstein Veblen, who propounded the doctrine of *conspicuous consumption*.
- According to Veblen, some consumers measure the utility of a commodity entirely by its price, *i.e.*, the greater the price of a commodity, the greater its utility.
- For example, diamonds are considered as a prestige good in society, and for the upper strata of the society the higher the price of diamonds, the higher the prestige value of them and therefore the greater utility or desirability of them.

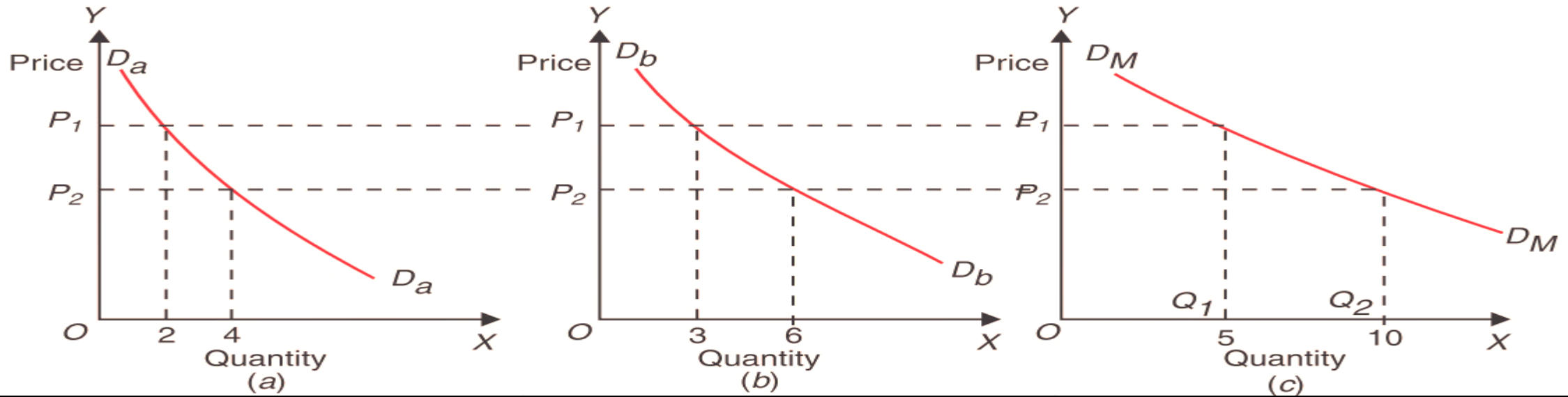
Giffen Goods

- Another exception to the law of demand was pointed out by Sir Robert Giffen.
- He observed that when price of bread increased, the low-paid British workers in the early 19th century purchased more bread and not less of it and this is contrary to the law of demand described earlier.
- After the name of Robert Giffen, such goods in whose case there is a direct price-demand relationship are called *Giffen goods*.
- *The demand curve will slope upward to the right and not downward.*

Market Demand Curve

- We can graphically obtain the market demand curve by making horizontal addition of the demand curves of all individuals buying the commodity.
- To do so, we add the various quantities demanded by the number of consumers in the market. In this way we can obtain the *market demand curve* for a commodity which, like the individual consumer's demand curve, will slope downward to the right.
- Suppose there are two individual buyers of a good in the market. Fig. (a) and (b) show the demand curves of the two independent individual buyers.
- Now, the market demand curve can be obtained by adding together the amounts of the goods which individuals wish to buy at each price. Thus, at price P_1 , the individual A wishes to buy 2 units of the goods; individual B wishes to buy 3 units of the goods.
- The total quantity of the goods that the two individuals plan to buy at price P_1 is therefore $2 + 3 = 5$, which is equal to OQ_1 .

Market Demand



	QD at P1	QD at P2
Consumer A	2	4
Consumer B	3	6
Market	5	10

Market Demand Function

- **Adding up Individual Demand Functions:** Demand for a commodity by an individual depends on the commodity's price, income, and related commodities' prices. A simple demand function is expressed in terms of own price of the commodity, keeping other factors constant. Thus, such a linear demand function is

$$Q_A = 40 - 2P$$

- where Q_A is the quantity demanded of the commodity by individual A, P is the price of the commodity in rupees, and 40 is constant (the intercept term). The above demand function shows that a rise in the price of the commodity by one rupee causes 2 units decrease in quantity demanded by individual A and vice-versa.
- Likewise, the demand functions of individuals B and C are as follows:

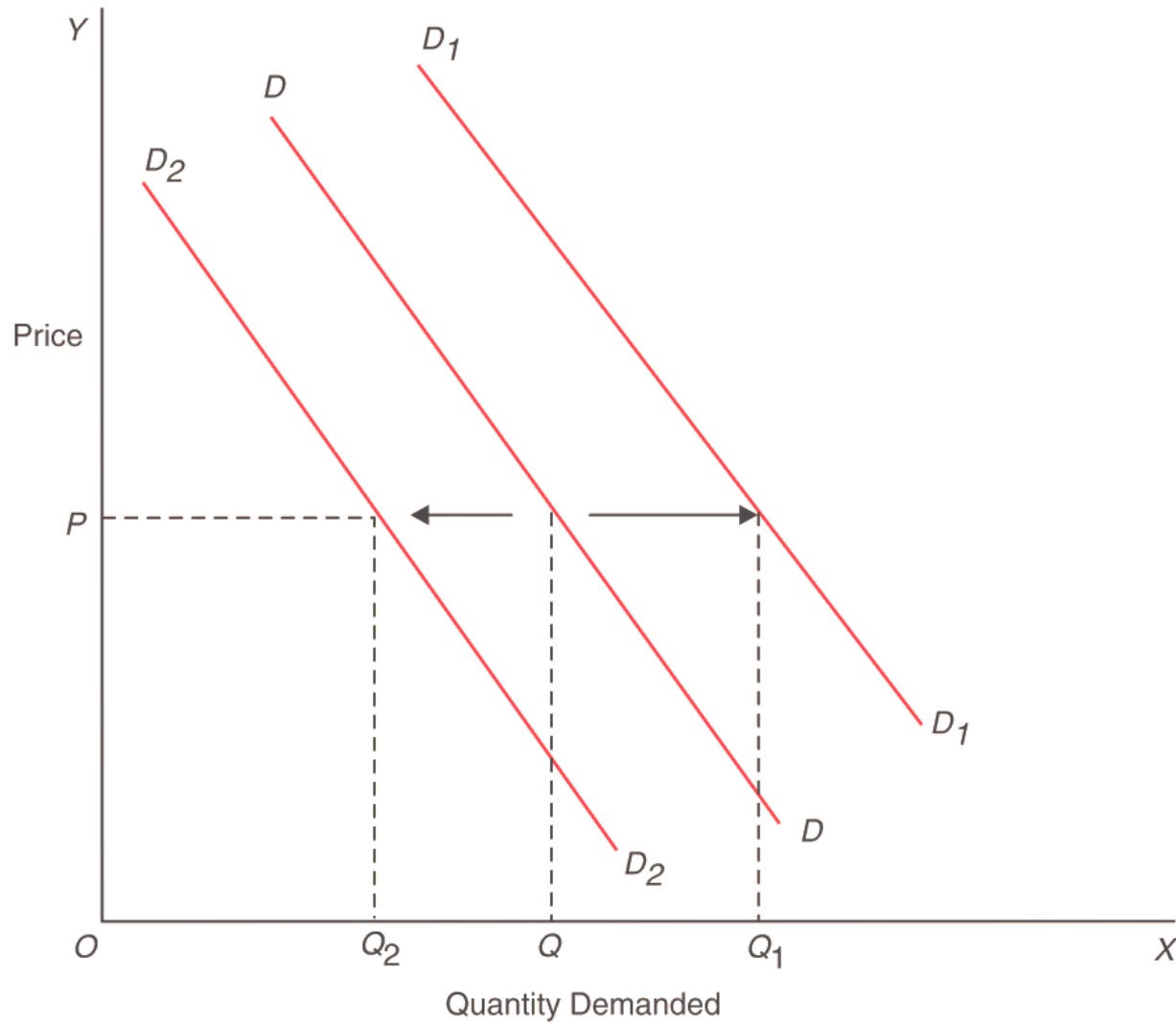
$$Q_B = 25.5 - 0.75P$$

$$Q_C = 36.5 - 1.25P$$

-
- The market demand (Q_m) is the *sum* of the demands of individuals who want to purchase the commodity.

- $$\begin{aligned} Q_a + Q_b + Q_c &= Q_m = (40.0 - 2P) + (25.5 - 0.75P) + (36.5 - 1.25P) \\ &= 40.0 + 25.5 + 36.5 - (2 + 0.75 + 1.25)P \\ &= 102 - 4P \end{aligned}$$

- If P is in rupees, the market demand function shows that a rise in the price of the commodity by one rupee will result in a fall in its quantity demanded by 4 units.



*Shifts in
Demand curve
due to
Changes in
Variables other
than Price*

To conclude, whereas demand function specifies relationship between quantity demanded of a product with many independent variables, demand curve of a product is a graphic representation of only a part of the demand function with price of the product as the only independent variable. The effects of change in variables other than price, as explained above, are shown through a shift in the demand curve.